Table: Geometrical properties of Bean and Cassini curves

Properties	Bean curves	Cassini ovals
Intercepts:	$(0,\ 0), (0,\ a_1^2)$	$(\pm\sqrt{a_2^2 \pm b_2^2},\ 0), (0,\ \pm\sqrt{b_2^2 - a_2^2})$
Extrema:	$(0,\ a_1^2), \left(\pm\sqrt{\frac{a_1^2 b_1^3(b_1-2a_1)}{16(b_1-a_1)^2}},\ \frac{a_1 b_1^2}{4(b_1-a_1)}\right)^*,$ $\left(\pm\sqrt{\frac{a_1^2 b_1^2}{16}(1+2\sqrt{\frac{a_1}{b_1}})},\ \frac{a_1 b_1}{4}\right)$	$\left(\pm\sqrt{a_2^2 \pm b_2^2}, 0\right), \left(0,\ \pm\sqrt{b_2^2 - a_2^2}\right)^*,$ $\left(\frac{\pm\sqrt{4a_2^4 - b_2^4}}{2a_2},\ \pm\frac{b_2^2}{2a_2}\right)$
Symmetries:	$x = 0;\ (0,\ 0)$	$x = 0;\ y = 0;\ (0, 0)$
Loops:	a single connected loop for $(\pm a_1,\ \pm b_1)$ three connected loops for $(\pm a_1,\ \mp b_1)$ —	a single loop if $a_2 < b_2$ the lemniscate of Bernoulli, if $a_2 = b_2$ two disconnected loops, if $a_2 > b_2$
Nodes:	$(0,\ \frac{a_1^2}{2})$ if $a_1 = b_1$	$(0, 0)$ if $a_2 = b_2$

Table: XYZ

σ	η	
	$Model(I)$	$Model(II)$
0.0	-5.12	-2.0
0.15	-5.04507	-1.92518
10.0	-10.3949	-7.62391
30.0	-22.2524	-21.375
70.0	-49.6698	-55.0179

Eigenvalues ω_0^2 and ω_1^2 of Eq.(26) for $r_0 = 1.1$ and varying B, for two string configuration

Apples					
A1		a1	B1		b1
Q1	Q2		Q1	Q2	
1	3	4	a	b	c
00	22	08	v	h	p